



Operations Research

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Solutions to the Demo Exercise, Sheet 10

Demo Exercises

Exercise D10.1. *Topological Properties and Convex Sets*

Problem

Consider the following two sets in $\mathbb{R}^2$:

$$X_1 = \{x \in \mathbb{R}^2 \mid \|x\|_2 < 1\}$$

$$X_2 = \{x \in \mathbb{R}^2 \mid x_1 + x_2 \leq 1, x_1 \geq 0, x_2 \geq 0\}$$

- (a) Determine for both X_1 and X_2 whether they are open, closed, bounded, and/or compact. Justify your answers mathematically.
- (b) Prove rigorously that X_2 is a convex set.

Solution

- (a) **Set X_1 :** This is an open Euclidean ball. It is **open** because for any $x \in X_1$, there exists an $\epsilon > 0$ such that $B_\epsilon(x) \subset X_1$. It is **bounded** because it is contained entirely within the closed ball $\bar{B}_1(0)$. It is **not closed** (as it does not contain its boundary $\|x\|_2 = 1$), and thus **not compact**.
Set X_2 : This set is **closed** because it is the intersection of three closed halfspaces. It is **bounded** because $x_1, x_2 \geq 0$ and $x_1 + x_2 \leq 1$ imply $x_1, x_2 \in [0, 1]$. By the Heine-Borel theorem, being closed and bounded in $\mathbb{R}^2$ means X_2 is **compact**. It is **not open**.
- (b) Let $x, y \in X_2$ and $\lambda \in [0, 1]$. We must show that $z = \lambda x + (1 - \lambda)y \in X_2$. Since $x_1, x_2, y_1, y_2 \geq 0$ and $\lambda, (1 - \lambda) \geq 0$, it immediately follows that $z_1 \geq 0$ and $z_2 \geq 0$. Furthermore:

$$\begin{aligned} z_1 + z_2 &= (\lambda x_1 + (1 - \lambda)y_1) + (\lambda x_2 + (1 - \lambda)y_2) \\ &= \lambda(x_1 + x_2) + (1 - \lambda)(y_1 + y_2) \end{aligned}$$

Since $x_1 + x_2 \leq 1$ and $y_1 + y_2 \leq 1$:

$$z_1 + z_2 \leq \lambda(1) + (1 - \lambda)(1) = 1$$

Thus, $z \in X_2$, proving X_2 is a convex set.

Exercise D10.2. Unconstrained Optimization

Problem

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be given by $f(x, y) = x^2y + y^3 - 3y$.

- (a) Compute the gradient $\nabla f(x, y)$ and the Hessian matrix $H_f(x, y)$.
- (b) Determine all stationary points of f .
- (c) Classify each stationary point (local minimum, local maximum, or saddle point) by explicitly calculating the eigenvalues of the Hessian matrix at each point.

Solution

- (a) The gradient is the vector of partial derivatives:

$$\nabla f(x, y) = \begin{pmatrix} 2xy \\ x^2 + 3y^2 - 3 \end{pmatrix}$$

The Hessian matrix contains the second-order partial derivatives:

$$H_f(x, y) = \begin{pmatrix} 2y & 2x \\ 2x & 6y \end{pmatrix}$$

- (b) Stationary points occur where $\nabla f(x, y) = 0$. From the first component: $2xy = 0 \implies x = 0$ or $y = 0$.

- **Case 1** ($x = 0$): Substitute into the second component: $0^2 + 3y^2 - 3 = 0 \implies 3y^2 = 3 \implies y = \pm 1$. This yields points $(0, 1)$ and $(0, -1)$.

- **Case 2** ($y = 0$): Substitute into the second component: $x^2 + 3(0)^2 - 3 = 0 \implies x^2 = 3 \implies x = \pm\sqrt{3}$. This yields points $(\sqrt{3}, 0)$ and $(-\sqrt{3}, 0)$.

This gives four stationary points: $(0, 1)$, $(0, -1)$, $(\sqrt{3}, 0)$, and $(-\sqrt{3}, 0)$.

- (c) We evaluate the eigenvalues λ of the Hessian by solving the characteristic equation $\det(H_f - \lambda I) = 0$:

- **At** $(0, 1)$: $H_f(0, 1) = \begin{pmatrix} 2 & 0 \\ 0 & 6 \end{pmatrix}$. As a diagonal matrix, the eigenvalues are simply $\lambda_1 = 2, \lambda_2 = 6$. Both are strictly positive ($H_f \succ 0$), making this a **Local Minimum**.

- **At** $(0, -1)$: $H_f(0, -1) = \begin{pmatrix} -2 & 0 \\ 0 & -6 \end{pmatrix}$. The eigenvalues are $\lambda_1 = -2, \lambda_2 = -6$. Both are strictly negative ($H_f \prec 0$), making this a **Local Maximum**.

- **At** $(\sqrt{3}, 0)$: $H_f(\sqrt{3}, 0) = \begin{pmatrix} 0 & 2\sqrt{3} \\ 2\sqrt{3} & 0 \end{pmatrix}$. The characteristic equation is:

$$\det \begin{pmatrix} -\lambda & 2\sqrt{3} \\ 2\sqrt{3} & -\lambda \end{pmatrix} = \lambda^2 - 12 = 0 \implies \lambda = \pm 2\sqrt{3}$$

Because the eigenvalues have mixed signs, the matrix is indefinite, making this a **Saddle Point**.

- **At $(-\sqrt{3}, 0)$:** $H_f(-\sqrt{3}, 0) = \begin{pmatrix} 0 & -2\sqrt{3} \\ -2\sqrt{3} & -\lambda \end{pmatrix}$. The characteristic equation is:

$$\det \begin{pmatrix} -\lambda & -2\sqrt{3} \\ -2\sqrt{3} & -\lambda \end{pmatrix} = \lambda^2 - 12 = 0 \implies \lambda = \pm 2\sqrt{3}$$

Again, mixed eigenvalues mean the matrix is indefinite, making this a **Saddle Point**.

Exercise D10.3. KKT-conditions

Problem

- (a) (4P) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}, g : \mathbb{R}^n \rightarrow \mathbb{R}^m, h : \mathbb{R}^n \rightarrow \mathbb{R}^p$.

Consider the following constrained non-linear optimization problem.

$$\min f(x) \text{ s.t. } g(x) \leq 0, h(x) = 0$$

State the KKT-conditions.

- (b) (1P) How do the KKT conditions change for a maximization problem?
You only need to write down the changes compared to (a).

For the following exercises let

$$\begin{aligned} f(x, y) &= \exp(2x^2 + 4y^2 + 24y) \\ h(x, y) &= x^2 + (y + 3)^2 - 1. \end{aligned}$$

And let the problem (P) be given by

$$(P) \quad \min f(x, y), \text{ s.t. } h(x, y) = 0$$

- (c) (3 P) Show that f is convex. You may use without proof that the composition $f_1 \circ f_2$ is convex if f_1 and f_2 are convex and f_1 is strictly increasing.
- (d) (1P) Show that Slater's condition is violated for (P).
- (e) (3P) Compute the gradients of f and h .
- (f) (2P) Show that $\nabla h \neq 0$ for all feasible points. This implies that all feasible points are regular.

You may assume without proof that the feasible region is compact, hence the continuous function f attains both a global minimum and maximum. Furthermore, assume that there exists four KKT-points in total (two for maximization and two for minimization), which are given by z_1, z_2, z_3 and z_4 .

- (g) (3P) Describe how you would determine the global minima and maxima of f under the constraint $h = 0$. Justify your approach.
You do not need to explicitly compute the maximizers and minimizers.

Solution

(a) • *Stationarity*: $\nabla f(x^*) + \sum_{i=1}^m \lambda_i \nabla g_i(x^*) \pm \sum_{j=1}^p \mu_j \nabla h_j(x^*) = 0$

• *Primal Feasibility*: $g_i(x^*) \leq 0, i \in [m], h_j(x^*) = 0, j \in [p]$

• *Dual Feasibility*: $\lambda_i \geq 0, i \in [m]$

• *Complementary Slackness*: $\lambda_i g_i(x^*) = 0, i \in [m]$.

(b) $\max f \Leftrightarrow \min -f$. The KKT conditions change accordingly.

(c) Note that $f_1 \circ f_2$ denotes the composition $f_1(f_2(x))$ and not the multiplication $f_1 f_2$.

$f_1(x) = \exp(x)$ is convex and strictly increasing: $f'_1 = f''_1 = f_1 > 0$.

$f_2 = 2x^2 + 4y^2 + 24y$ is convex, since its Hessian is a diagonal matrix with positive entries. $H_{f_2} = \text{diag}(4, 8)$

Alternatively, we compute the hessian of f directly:

$$H_f(x, y) = f(x, y) \underbrace{\begin{pmatrix} 4 + 16x^2 & 32x(y+3) \\ 32x(y+3) & 8 + 64(y+3)^2 \end{pmatrix}}_A$$

Since H_f is symmetric we can use the principal minor criterion to determine the definiteness of H_f . Since $f > 0$, $4 + 16x^2 > 0$ we need to show that $\det(A) > 0$.

$$\begin{aligned} \det(A) &= (4 + 16x^2)(8 + 64(y+3)^2) && -32^2 x^2 (y+3)^2 \\ &> (16x^2)(64(y+3)^2) && -32^2 x^2 (y+3)^2 = 0. \end{aligned}$$

(d) Since the equality constraint is not affine, Slater's condition does not hold.

(e)

$$\nabla f = f(x, y) \begin{pmatrix} 4x \\ 8(y+3) \end{pmatrix}, \nabla h = \begin{pmatrix} 2x \\ 2(y+3) \end{pmatrix}$$

(f)

$$\nabla h = 0 \Leftrightarrow \begin{pmatrix} 2x \\ 2(y+3) \end{pmatrix} = 0 \Leftrightarrow x = 0, y = -3 \Rightarrow h(x, y) \neq 0$$

(g) The KKT-conditions are necessary for optimality, since all points are regular by the previous exercise. Global extrema are local extrema. Since a global minimum/maximum exists there exists an optimal KKT-point. Comparing the function values of the KKT-points z_i yields the corresponding extrema.

Note that the hessian of the objective does not provide any information for the type of an extremum under constraints. Consider the following example:

$$\min x^2 \text{ s.t. } 0 \leq x \leq 1$$

Then $x^* = 1$ is a global maximizer, but $H_f = 2 > 0$ is positive definite.